Project Milestone 02

Objective:

The third milestone of the project is with the following details:

Milestone 02	Date	Description
M-02 (**6%**)	**Tuesday 17th of March, 2026**	1. Hardware assembly, circuit and connections. 2. Acquire the images from the **external camera** connected to the **Raspberry Pi** (Use the Raspberry Pi camera or a USB camera connected to the Pi) connected to the low-level microcontroller **(Arduino)** for motor actuation. 3. Implement and run **Geometric Transformation, Intensity Level Transformation and Image Smoothing codes** on the acquired video **by your hardware (Choose the appropriate algorithm for your project)**. 4. **Simple closed loop** implementation to enable the motor to take an action depending on the image features (ex. Brightness, Rotation, etc.). 5. **Compare** the **Pi** and the **Laptop** performances, performing the same operations using the laptop instead of the Pi on the assembled hardware.

**The weight of each deliverable is presented between brackets beside the name of each deliverable.*

Notes: ***EVALUATIONS for this milestone will take place on Tuesday 17th of March, 2026 to see all your hardware and the image processing operations and closed loop system requirements implemented on the hardware so be ready =). Evaluations schedule and location will be announced soon.***

Requirements:

The requirements from this milestone of the project are as follows:

1. Each team is required to provide a fully fabricated hardware and circuit building, in addition to all the achieved connections. The hardware should have the camera(s) fixed on the system with all the wiring ready to acquire images from the surrounding for processing and actuating the motor accordingly. ***Note that each team has to consider the Raspberry Pi in the control loop as the main decision maker having the image processing code running and sending the control action to a microcontroller (Arduino) to actuate the system.***
2. For the Raspberry Pi connections, you can use either the **Pi camera or an external USB camera** connected to the Pi to acquire the image and perform the image processing operations on the Raspberry Pi. (Note that you have to make sure that the used camera is suitable for Pi connection). Each team should **connect the Raspberry Pi to an Arduino** to send the required commands and the Arduino should be the one responsible in actuating the motors to avoid any harm that can be caused to the Raspberry Pi.

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3. Each team is required to **acquire images/video from the camera connected to the Pi** and perform several image processing techniques on the system. These processing techniques should include:
    a. **Two geometric transformation** operation (ex.: rotation, scaling, transformation, etc.)
    b. **Two intensity level transformation** operations (ex.: brightening, darkening, contrast enhancement, etc.)
    c. **One smoothing algorithm** (ex.: averaging filter, gaussian filter, etc.)

    to transform and filter the acquired image and try to extract from it the desirable information for your assigned system to function appropriately. (Note the codes should be written in **Python** and **should be executed from the Pi**).
4. Each team is required to perform a **simple closed loop** implementation for the system to enable the motor to **take an action depending on the image features extracted** (ex. If the image is bright then rotate the motor(s) clockwise with a changing speed depending on the brightness percentage of the image, If the image is dark then rotate the motor(s) counter-clockwise with a changing speed depending on the darkness percentage of the image **or** if the edges counted are more than a threshold then rotate the motor(s) clockwise, else rotate the motor(s) counter-clock wise **or** others (**try to be creative and project specific as these codes will help you performing the required task of the project in the upcoming milestones =)**)). This task should be implemented using the live streaming technique by which the image/video is acquired, processed then the motor should operate accordingly. Each team **MUST choose TWO different image features** (**two different scopes; brightening and edge detection NOT brightening and darkening for example**) coded as if conditions to be applied for the motor control. It is preferable to choose the features that are most applicable with your application.
5. Each team should execute the same image processing codes and closed loop experiment codes (mentioned in requirements 3 and 4) connecting the hardware system to the **Laptop instead of the Raspberry Pi** and **compare** their performance in terms of **execution time, accuracy and complexity**.

Submission:

Submit the requirements through the following link: https://docs.google.com/forms/d/e/1FAIpQLScxl7ANL6zCBJOM3BXnX0gPISiFC20pO5Uy6pJ_555_27e5Zg/viewform?usp=publish-editor. Name the folder added to your submitted link as: ***"Milestone_02_Team_#"***. The folder should include the following:

The contents of the zip file:

1. **ZIP file** includes the following items:
    - For teams using **Python** as the coding language, you have to submit the Python scripts **for this milestone only** (if any) saved in **(.ipynb).**

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- For teams using **MATLAB** as the coding language, you have to submit the MATLAB scripts **for this milestone only** (if any) saved in **(.m)** and MATLAB workspace saved in **(.mat).**
- ALL teams should save the images or videos acquired by the camera, in addition to the processed images that contribute to the decision making of the image.

2. **ZIP file** includes the following items:
   - Any updates done (if any) in the previously submitted SOLIDOWRKs design or circuit design as the final design for the overall system.
3. **Word file** that includes the following. **Build on the previously submitted report.**
   - The report should include the overall hardware fabricated, circuit built and all the connections done with the Raspberry Pi/microcontrollers.
   - This report should also include the image features used to actuate the motor and the response figures of the motor response indicating the varying speed and direction of the motor rotation. You should comment on the results obtained relating the image features and motor response.
   - You should include the comparison between the Laptop and Pi in the system's performance.
4. **PDF file** including a PDF version of the whole report including all the previously mentioned tasks.
5. **Narrated videos** that include the following:
   - **Video 1:** This video should include all the previously implemented operations in Milestone 02 acquired by the fixed camera.
   - **Video 2:** This video should include the hardware fabricated and circuit connected illustrating the system, showing the simple closed loop response done using the defined image features.
   - **Video 3:** This video includes the comparison between the Laptop and the Pi.

The deadline for the submission is **Tuesday 17th of March at 11:59 PM.** Late Submissions will result in deduction from the grade of this deliverable.

Be Ready for EVALUATIONS for this milestone that will take place on Tuesday 17th of March, 2026.
Evaluations schedule and location will be announced soon.